Evaluation Document of DAugtify – An Automated Data Augmentation Extension for Image Classification Model Training Pipeline

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Note

This document was created to give the reader an initial impression of the project such that the final form (Section "Final evaluation of the project") can be filled. This is not the final document.

Introduction

The performance of modern machine learning models heavily depends on the quality and quantity of the training dataset. However, collecting proper datasets is costly and labor-intensive in many domains such as medical and agricultural.

Data augmentation is an effective and efficient technique to ease this issue, which aims to improve poor datasets by creating new variations using existing data. A few examples of traditional data augmentation techniques include flipping images horizontally or vertically, rotating images at different angles, and changing the color of images.

Figure 1: Flipping technique where (a) original image, (b) vertical flipping, (c) horizontal flipping, and (d) vertical and horizontal flipping

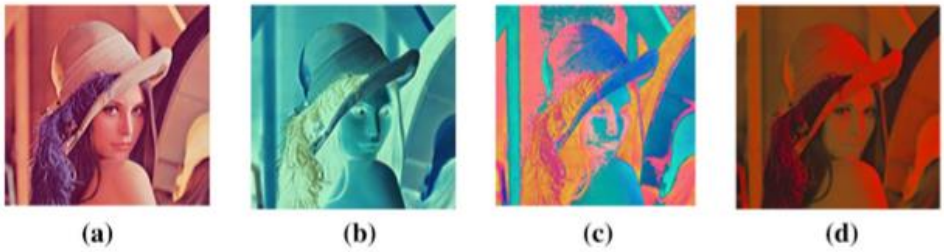

Figure 2: Color space for (a) RGB, (b) CMY, (c) HSL, and (d) YIQ

Though data augmentation is a good solution to improve poor diverse datasets, not all data augmentation techniques are suitable for every dataset because each dataset has unique characteristics. Hence, selecting an optimal data augmentation scheme based on the given dataset and task is time-consuming and requires domain expertise.

In recent years, there have been some efforts to automate the selection of the optimal data augmentation scheme based on the given dataset task. These techniques are named Automated Data Augmentation (AutoDA). However, these research works have serious limitations such as **high computational usage and complexity of implementation**, which make them impractical for real-world settings. For example, AutoAugment requires **5000** NVIDIA Tesla P100 GPU hours to perform AutoDA tasks on the CIFAR10 dataset (Cubuk et al., 2019a). Later works on the domain were able to reduce this time by using complex implementations and powerful GPU resources (Yang et al., 2022).

Considering these problems, the author aims to bridge this gap by introducing **a lightweight single hyper-parameter neural network that can learn optimal data augmentation schemes simultaneously with machine learning model training.** During the initial stage, I only considered image classification model optimization with automated data augmentation.

Importance of the project

Data augmentation is a widely accepted solution to improve machine learning model performance. Automating this process can save time and resources while producing better results. Hence, automated data augmentation models (**AutoDA models**) have a high potential to become an important part of standard machine learning model training pipelines in the future.

How did I solve the problem?

Inspired by the findings of the Google Brain team in their work on RandAugment (Cubuk et al., 2019b), I incorporated the concept of shared magnitude into my solution approach to address the problem.

1. Redefinition of search space

Search space of AutoDA models refers to the set of image transformation techniques, which are: contrast, equalize, rotate, solarize, color, posterize, brightness, sharpness, shear, and translate. Typical AutoDA search space consists of 3 major parameters,

- **P:** Probability of applying each transformation
- **M:** Magnitude for each transformation
    - For example: Rotate by 9 degrees, in this example 9 is the magnitude.

- **N:** Number of augmentation operations to apply sequentially

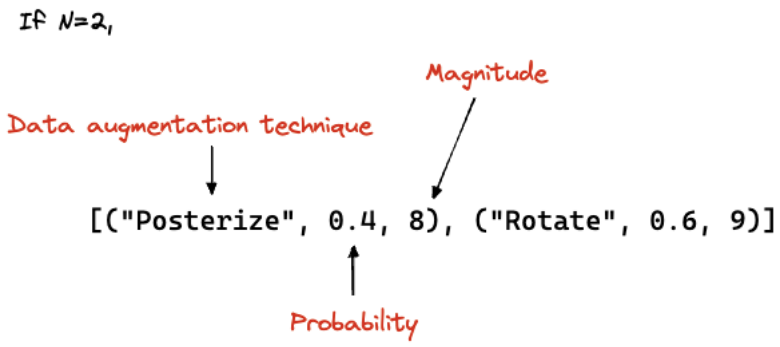

Figure 3: Example Data Augmentation Policy

However, after conducting a series of experiments and a thorough literature review on data augmentation I have reparametrized the search space to a single parameter. The proposed search space consists of:

- **P: Equal Probability for all available techniques**
  - If the search space consists of 14 image transformation techniques, the probability for each transformation will be 1/14.
- **M: Shared magnitude for all image transformations**
  - Instead of searching for a specific magnitude for each transformation, the system now looks for a common magnitude that works for all image transformations. Additionally, the effective range for each image transformation has been established, and when a relevant transformation is performed on the image, the shared magnitude is mapped to the effective range. Ultimately, this eliminates the need to search for a specific magnitude for each transformation.
- **N: The value N will always be 2**
  - Experiments that have been conducted throughout the development have proved that applying more than 2 image transformation operations sequentially can damage the characteristics of the image.

2. Development of a single hyper-parameter neural network

To determine the optimal magnitude for all image transformation techniques, a single hyper-parameter neural network was introduced, which represents the shared magnitude for all image transformations. This neural network is trained simultaneously with the image classification model, and it is capable of randomly applying 2 image operations to the training images using

the shared magnitude. The shared magnitude is optimized based on the loss value of the image classification model. Furthermore, an early stop callback has been implemented for this neural network, which halts training if the shared magnitude remains unchanged for three consecutive epochs. The optimal magnitude at that point is then used to augment the images for the remaining epochs.

The redefinition of the AutoDA search space involved reducing the number of possible data augmentation scheme combinations and hyperparameter tuning. This led to significant time savings during the AutoDA process. Moreover, the introduction of the lightweight single hypermeter neural network concept further optimized the search space exploration strategy. By using a single hyperparameter neural network, the exploration process became less complex, and the computational resources required for the task were significantly reduced. Taken together, these changes have greatly improved the efficiency of the AutoDA process by reducing the **search time and complexity and the computational resources** required for performing the AutoDA task.

Demonstration

The requirements gathered during the requirement gathering phase of this project have proven that this system should be developed as a Python library. However, to fulfill the requirements of the University of Westminster I have developed a web application for demonstration purposes.

Note: The training time for the image classification model can vary based on the hyper-parameter amount of the classification model and epoch count. Therefore, this demonstration has been limited to a single epoch to showcase the functionalities of the system.

The demonstration video **(<2 mins**) can be found @ Google Drive Link

Testing Criteria & Results

The testing of the system was conducted with various benchmarking datasets and image classification models. Below I have described one of them.

Dataset: CIFAR10

Image classification model: MobileNetV2 (**NOT** pre-trained, 3.2M params)

Epoch: 10

Hardware configurations: Apple MacBook Pro 2021 – M1 Pro chip, 16GB RAM

Training Accelerator: None

Criteria	Accuracy (%)	Time taken for Training
No augmentation	68.8	16h 50min
AutoAugment	70.5	N/A
RandAugment	72.2	N/A
DAugtify (proposed)	75.6	18h 30min

Table 1: Testing Results

Upon examination of the results, it has been demonstrated that the proposed DAugtify method archived higher accuracy while maintaining a competitive total training time compared to other methods.

To clarify, the results of training a MobileNetV2 model without augmentation achieved an accuracy of 68.8% in 16 hours and 50 minutes, while training the same model with the proposed AutoDA system achieved an accuracy of 80% in 18 hours 15 minutes. Although the training time with the proposed AutoDA was a little longer, the improvement in accuracy is significant enough to justify the additional training time.

Note: The PyTorch official implementation of AutoAugment and RandAugment provides pre-defined data augmentation schemes, rather than performing an actual search process (PyTorch implementation). Therefore, the model training time with AutoAugment and RandAugment AutoDA methods cannot be considered when doing a fair comparison.

The proposed DAugtify system was evaluated on an Apple MacBook Pro 2021 with an M1 Pro chip and 16GB of RAM. This demonstrates that DAugtify can operate without the need for high-end GPU resources, unlike existing methods that are dependent on such resources.

Future enhancements

- Automatically handle the class imbalance issue if present in the dataset.
- Improve the search space with recent image data augmentation techniques.

- Measure the effectiveness of other computer vision tasks like image segmentation and object detection.
- Create an open-sourced library for widespread use & improvement.

Contributions to the Body of Knowledge

The contributions made by completing this project to the existing body of knowledge are a novel resource-friendly solution to perform automated data augmentation.

- Redefinition of traditional AutoDA search space and prove the effectiveness.
- Development of resource-friendly lightweight single hyper-parameter neural network to search for optimal shared magnitude value for all image transformations and prove the effectiveness.
- Altogether novel approach to perform AutoDA task with reduced computational power and complexity while significantly improving the classification model performance.

Final evaluation of the project

By reading through the specifics of DAugtify, I hope you got insights as to what problem I am attempting to overcome and how it has been solved. I would highly appreciate it if you could take your valuable time to fill out an evaluation form on my project.

The evaluation form can be found @ Google Form Link

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Cubuk, E.D., Zoph, B., Mané, D., Vasudevan, V., Le, Q.V., 2019a. AutoAugment: Learning Augmentation Strategies From Data, in: 2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). Presented at the 2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 113–123. https://doi.org/10.1109/CVPR.2019.00020

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